

Introduction to Databases

Autumn 2025

Exercise 2

(Theory) Hand-in: 22.10.2025 (ADAM, 23:59)
(Practical) Hand-in: 22.10.2025 (during exercise)

Solving the Exercises: The exercises can be solved in small groups of a maximum of two people. Use the notations introduced in the lecture. The DMI plagiarism guidelines apply for this lecture.

Submission Information: Please upload all (Theory) deliverables BEFORE the deadline to ADAM as a **single PDF**. For (Practical) exercise check the **Hand-In** instructions. The solutions must contain your names. Solutions that are handed in too late cannot be considered.

Task (Practical) 1: Connecting to a Database (3 Points)

This week you will do the same SQL queries as last week — but this time with Python.

To connect to PostgreSQL using Python, you first need a so-called driver. This driver knows how to talk to PostgreSQL, and translates between Python and PostgreSQL types. For Python you can use the `psycopg` driver.

After installing the driver you need to connect to the database. Use the same connection details as for the last exercise. Once connected you need a *cursor*. This cursor can then be used to *execute* statements and retrieve the results.

On ADAM there is an example script, which prints the version of the database.

Use the script on ADAM as basis for your solution. Extend it to solve the following tasks. Use **only the SQL queries from last week**.

- Avoid hard-coding the password in the source file (e.g. prompt the user or use environment variables).
- Print how many games were won by white.
- Print the number of games per month. Print the name of the month, not the number.
- Print the total number of games.

If you are unfamiliar with Python, we recommend to use the `uv` tool¹. The script can be executed like this: `uv run ex2.py`. We will give a short demonstration on how to use it

¹<https://github.com/astral-sh/uv/releases>

in the exercise session on the 8th of October. If you have trouble with the setup, feel free to approach us during an exercise session.

Hand-In: Show your code and the printed results to an assistant/tutor.

Task (Theory) 2: Extended Entity Relationship Diagrams (7 Points)

Expanding its platform capabilities, the company behind FoodFusion now also wants to have a detailed management of driver logistics and vehicle usage for its delivery services. The detailed requirements for the data model are the following:

Drivers are stored with their first and last name, the license number, and contact information.

Vehicles are stored with a vehicle ID (**VID**) and common attributes such as license plate and carrying capacity. There are different types of vehicles: *cars*, *scooters*, and *bikes*. Each vehicle belongs to exactly one of these types. Cars additionally store the number of doors and trunk volume; scooters store information about battery capacity and electric range.

The system keeps track of the *assignment* of driver to vehicle together with a start and end time. A vehicle may only be assigned to one driver at any given time, while a driver may use different vehicles over time.

For cars only, a *fuel log* records the date, amount of fuel filled, and the cost. Each log entry is associated with a single car, but each car can have multiple log entries.

Deliveries are stored with pickup and delivery locations, time of order placement, priority level, and delivery status. Each delivery is fulfilled by exactly one driver and one vehicle.

Model the above circumstances by means of an Extended Entity Relationship diagram. Try to convert the requirements as precisely as possible (however, make sure that the EERD is still useful in the given setting, i.e., insert if necessary missing fields which may not be specified clearly in the requirements, but which you deem necessary). Ensure to mark primary keys and add, where it seems necessary to you, artificial keys. Make sure to add cardinalities to all relationships. Describe the requirements that cannot be represented in the Extended Entity Relationship diagram.

Make sure you write legibly and present your Extended Entity Relationship diagram in a clear way. Use the notation introduced in the lecture.

Hand-In: Upload your solution to ADAM.